DEPARTMENT OF COMPUTER SCIENCE

Course Title: Ethical Hacking

Robtex — Network and DNS Intelligence Tool

Objective

The objective of this experiment is to use Robtex, a free web-based network and DNS intelligence tool, to perform passive footprinting of a target domain. Students will learn to map DNS records, discover related domains sharing infrastructure, and identify network/ASN ownership information, entirely through Robtex's public web interface.

Methodology

1. Introduce Robtex as a passive, browser-based DNS and network intelligence tool.
2. Perform a domain lookup to review DNS records (A, MX, NS, TXT).
3. Use Robtex's graph view to visualise relationships between a domain, its IP addresses, and shared-hosting neighbours.
4. Perform an IP address lookup to identify the network range and ASN (Autonomous System Number) it belongs to.
5. Cross-reference Robtex findings with tools used in earlier weeks (WHOIS, crt.sh, DNSdumpster).

Expected Outcome

By completing this experiment, you will be able to:

1. Explain what Robtex is and how it fits alongside other passive footprinting tools.
2. Perform a domain lookup and interpret the returned DNS records.
3. Use Robtex's graph view to identify shared-hosting relationships.
4. Look up an IP address to identify its network range and ASN.
5. Cross-reference Robtex data against other OSINT sources for verification.

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1. Introduction to Robtex

Robtex (robtex.com) is a free, web-based network and DNS intelligence tool. It aggregates publicly available DNS, routing, and network registration data into a single, browsable interface, including an interactive graph that visualises how a domain, its IP addresses, and neighbouring domains on the same server relate to one another.

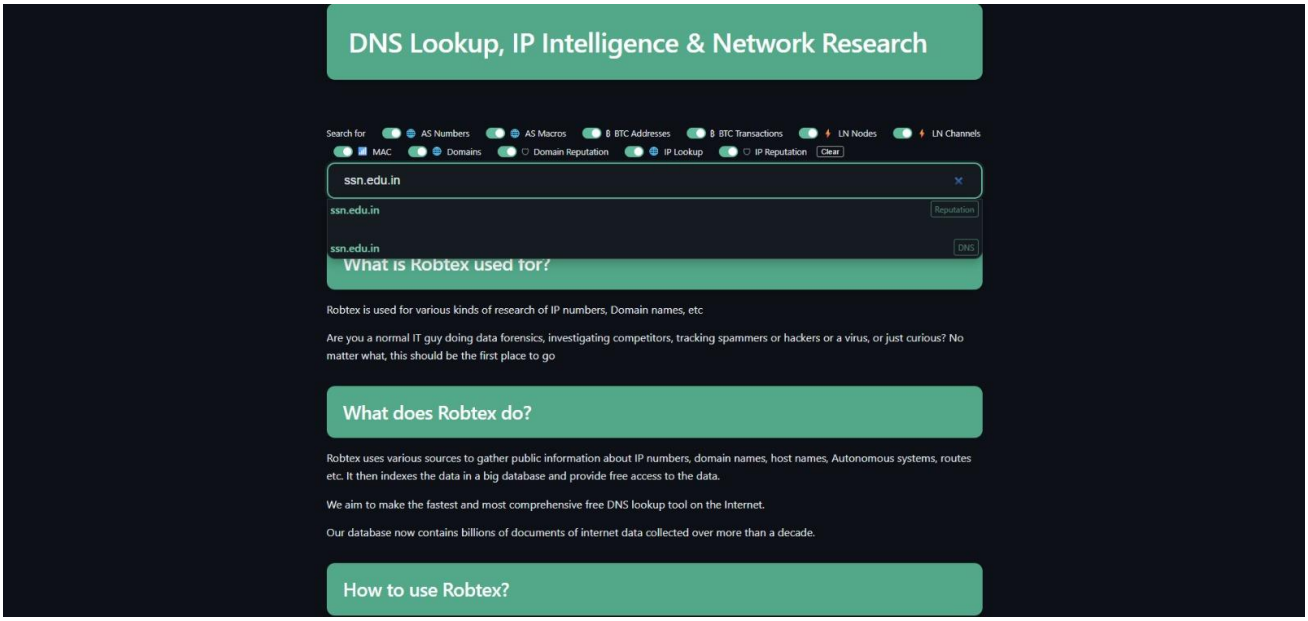
1.1. Why It Matters

1. Robtex is entirely browser-based, similar to DNSdumpster.
2. Its graph view often reveals shared-hosting relationships that a plain WHOIS or nslookup query would not show.
3. Purely passive — only queries Robtex's own aggregated public data, never the target's infrastructure directly.

2. Accessing Robtex

Step 1: Open Robtex in a Browser

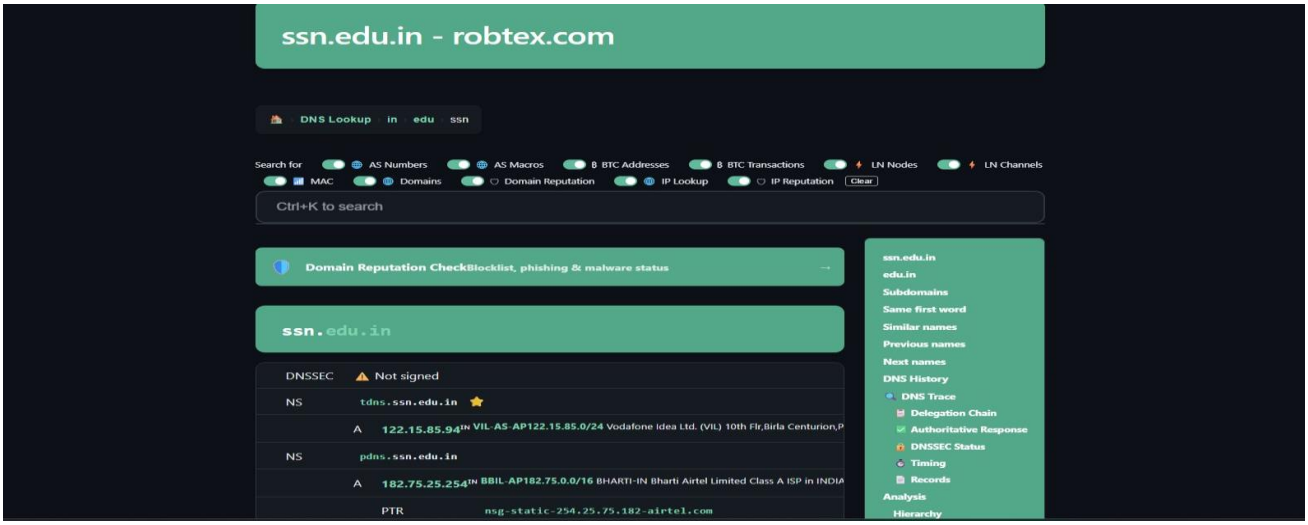
Navigate to https://www.robtex.com in any standard web browser. No account or installation is required.



3. Domain Lookup

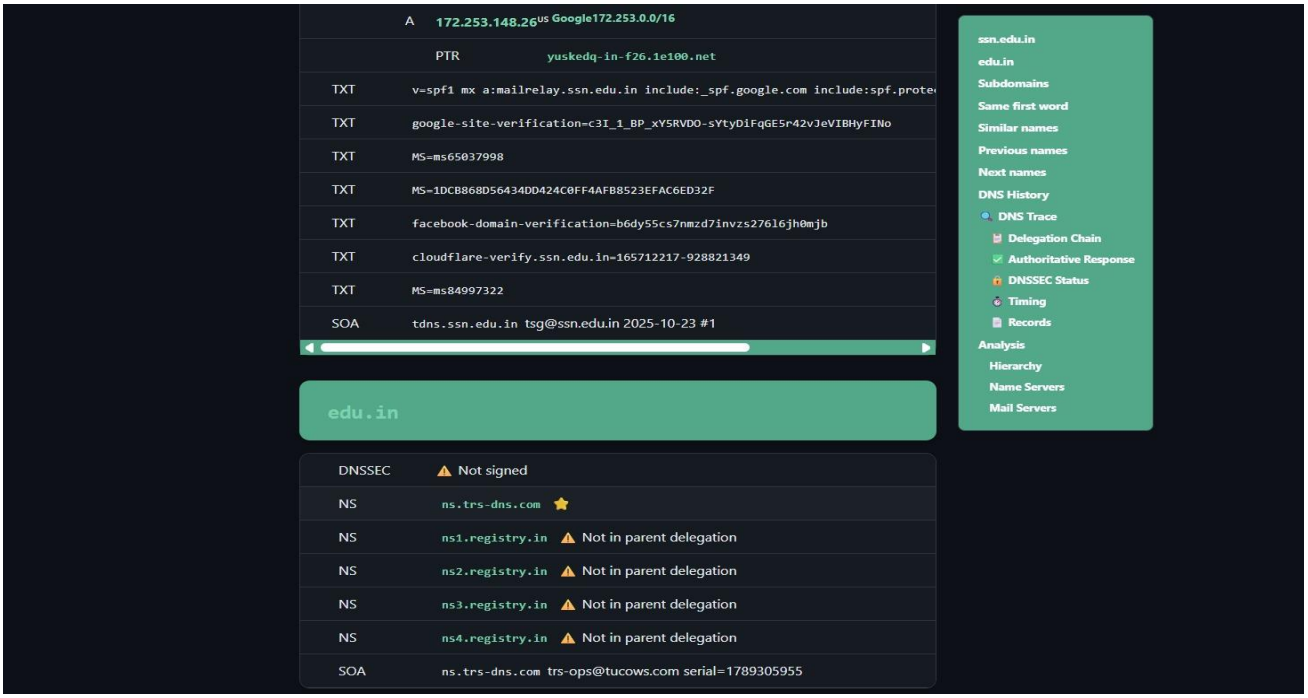
Step 1: Enter the Target Domain

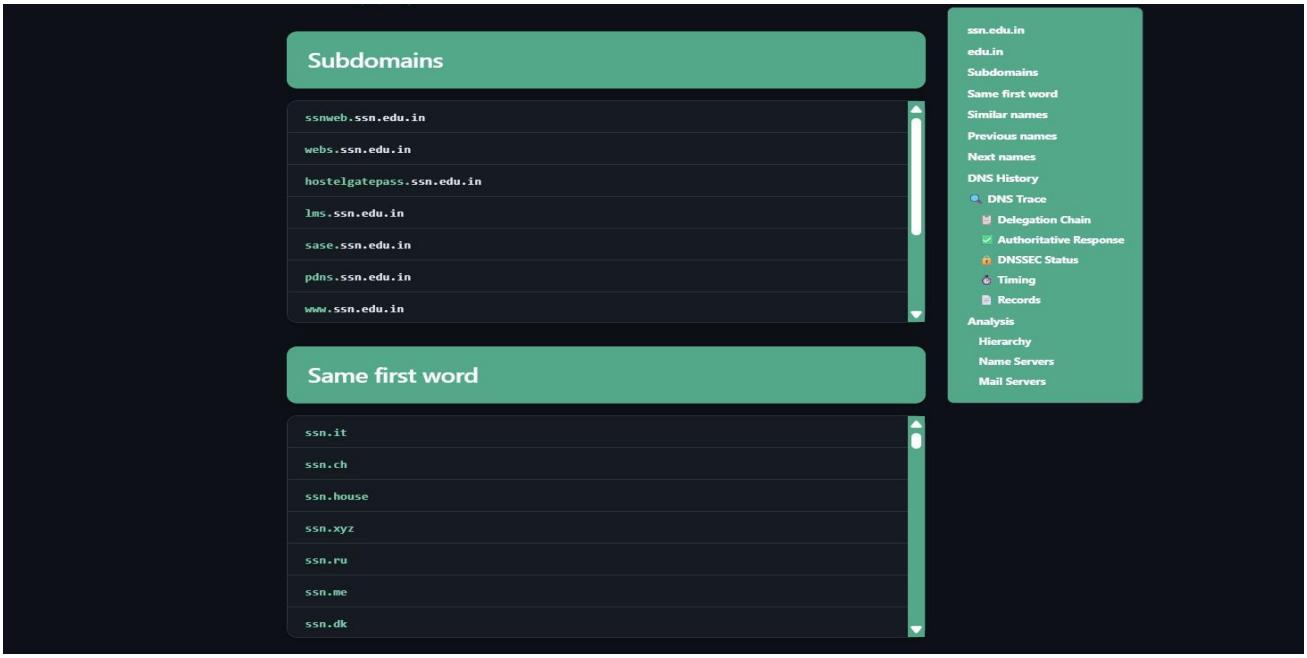
Type your authorised training domain into the Robtex search bar and press Enter.



Step 2: Review the Summary Panel

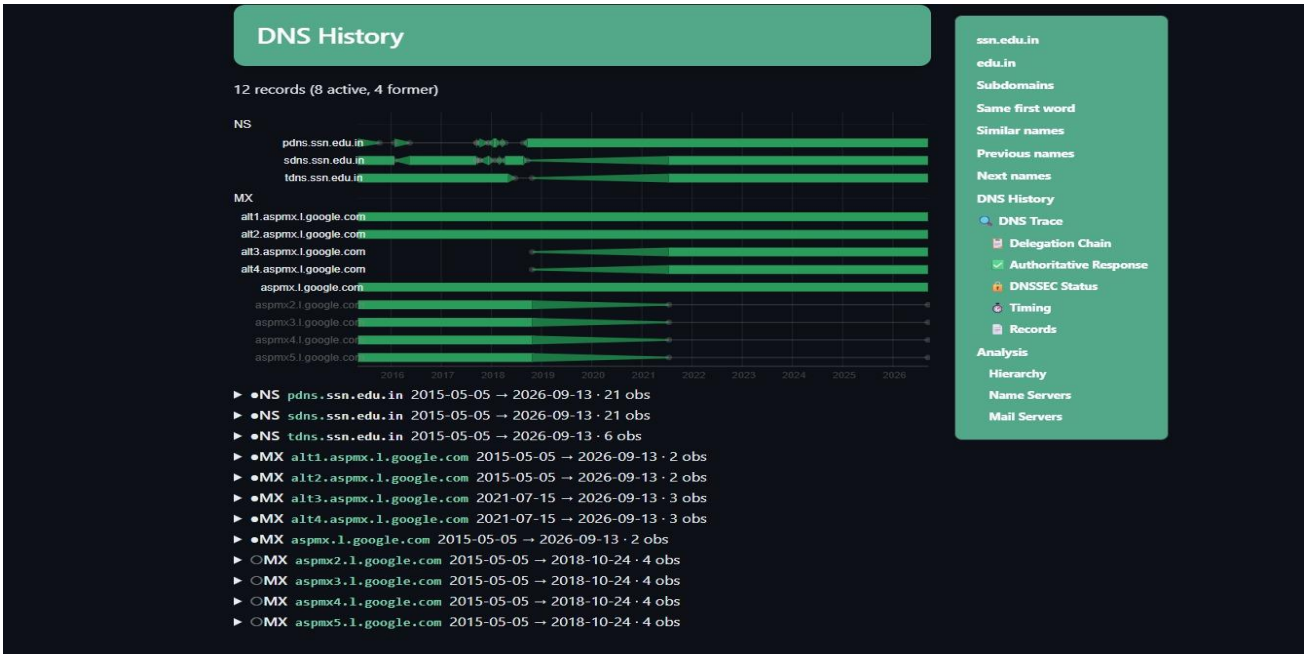
Review the initial summary showing the domain's resolved IP address(es) and hosting details.

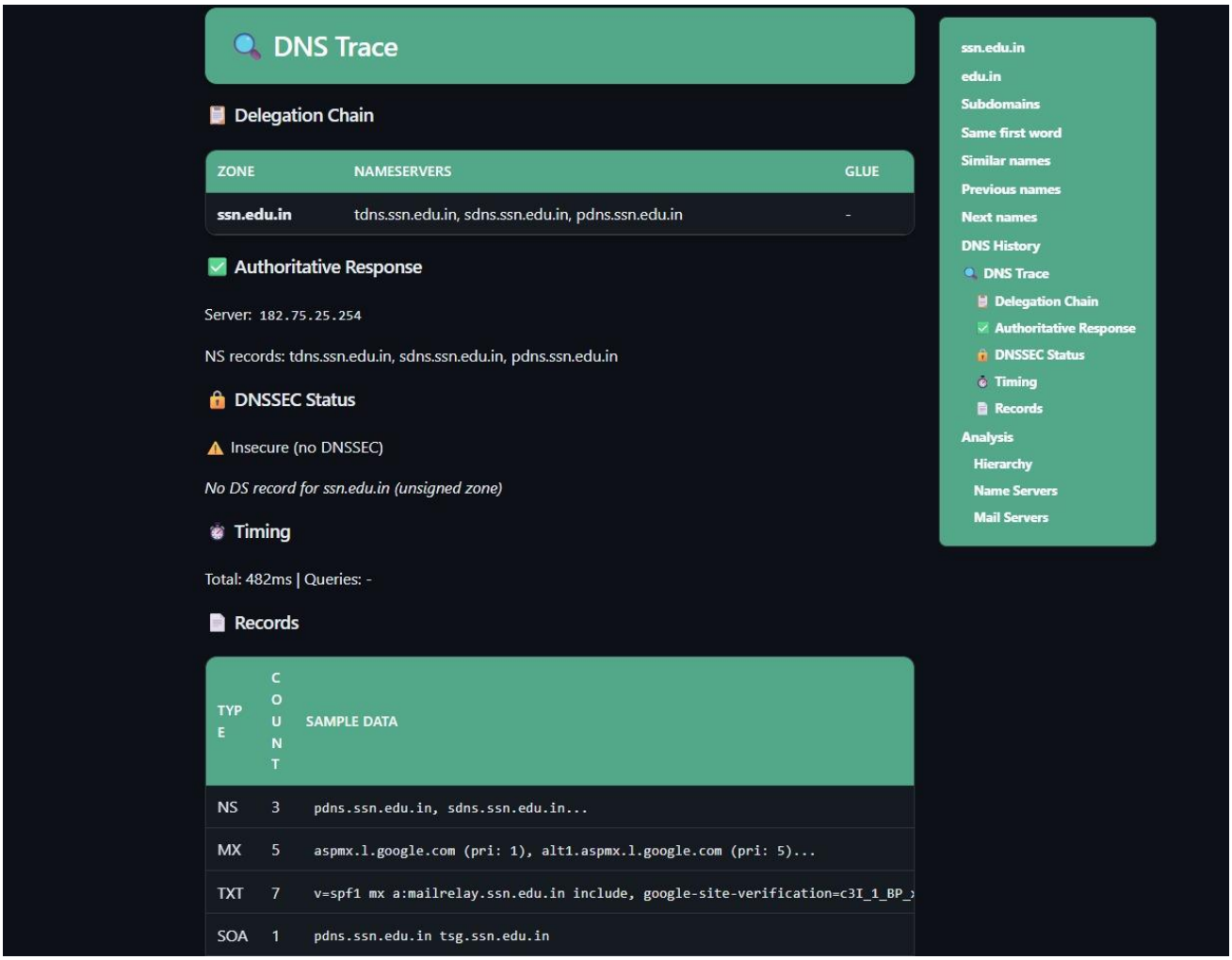




4. Interpreting DNS Records

1. A record — maps the domain to an IPv4 address.
2. MX record — identifies the mail server(s) responsible for the domain's email.
3. NS record — identifies the domain's authoritative name servers.
4. TXT record — often contains SPF/DKIM policies or domain verification strings.




DNS Trace
Delegation Chain
ZONE
NAMESERVERS
GLUE
ssn.edu.in
tdns.ssn.edu.in, sdns.ssn.edu.in, pdns.ssn.edu.in
-
Authoritative Response
Server: 182.75.25.254
NS records: tdns.ssn.edu.in, sdns.ssn.edu.in, pdns.ssn.edu.in
DNSSEC Status
Insecure (no DNSSEC)
No DS record for ssn.edu.in (unsigned zone)
Timing
Total: 482ms | Queries: -
Records
TYPE
COUNT
SAMPLE DATA
NS 3 pdns.ssn.edu.in, sdns.ssn.edu.in...
MX 5 aspmx.l.google.com (pri: 1), alt1.aspmx.l.google.com (pri: 5)...
TXT 7 v=spf1 mx a:mailrelay.ssn.edu.in include, google-site-verification=c3I_1_BP_
SOA 1 pdns.ssn.edu.in tsg.ssn.edu.in
ssn.edu.in
edu.in
Subdomains
Same first word
Similar names
Previous names
Next names
DNS History
DNS Trace
Delegation Chain
Authoritative Response
DNSSEC Status
Timing
Records
Analysis
Hierarchy
Name Servers
Mail Servers